AN ENTROPY BOUND FOR THE ATTRACTIVE LOGARITHMIC GAS

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Abstract. Assuming a logarithmic Hardy–Littlewood–Sobolev inequality, we prove a bound on the attractive logarithmic partition function that is uniform in the number of particles. A positive-definite cutoff separates a smooth bulk, which inherits the entropy inequality, from a singular interaction supported near the diagonal. The latter is perturbative when each box contains few particles; the complementary event is exponentially small. We give the required estimate for centered arrays of independent variables and include a cell-projection proof of the continuous-kernel bound in an appendix.

1. Setting and partition bound

Let $\mathsf{d} \geq 2$ and let μ_{unif} be normalized Lebesgue measure on $\mathbb{T}^{\mathsf{d}} = \mathbb{R}^{\mathsf{d}}/\mathbb{Z}^{\mathsf{d}}$. The mean-zero periodic logarithmic kernel is normalized by

$$\widehat{\mathsf{g}}(k) = \frac{c_{\mathsf{d}}}{(2\pi|k|)^{\mathsf{d}}} \quad (k \neq 0), \qquad c_{\mathsf{d}} = 2^{\mathsf{d}-1}\pi^{\mathsf{d}/2}\Gamma(\mathsf{d}/2), \qquad \mathsf{g}(x) = -\log|x| + O_{\mathsf{d}}(1) \quad (x \to 0). \tag{1.1}$$

Here, $\widehat{f}(k) = \int f(x)e^{-2\pi ik\cdot x}\,\mathrm{d}\mu_{\mathrm{unif}}(x)$ and $\widehat{\mathsf{g}}(0) = 0$. For independent $X_1, \dots, X_N$ with law μ_{unif}, set

$$\mathsf{W}_N(K) \coloneqq \frac{1}{N}\sum_{i\neq j} K(X_i, X_j), \qquad Z_{N,\beta} \coloneqq \mathbb{E}\big[e^{\frac{\beta}{2}\mathsf{W}_N(\mathsf{g})}\big], \tag{1.2}$$

where a function of one variable denotes the translation-invariant kernel $K(x-y)$.

Our only macroscopic assumption is the log-HLS inequality: for some $B \in (0, 2\mathsf{d}]$,

$$\mathsf{H}(\rho\mu_{\mathrm{unif}} \mid \mu_{\mathrm{unif}}) \coloneqq \int \rho\log\rho\,\mathrm{d}\mu_{\mathrm{unif}} \geq \frac{B}{2}\iint \mathsf{g}(x-y)(\rho(x)-1)(\rho(y)-1)\,\mathrm{d}\mu_{\mathrm{unif}}(x)\,\mathrm{d}\mu_{\mathrm{unif}}(y) \tag{1.3}$$

for every probability density of finite entropy. No value or optimality of B is needed.

Theorem 1.1 (Uniform partition bound)**.** *Under* (1.3)*, for every* $0 \leq \beta_0 < B$ *there is a finite constant* $C = C(\mathsf{d}, B, \beta_0)$ *such that*

$$\sup_{N\geq 2}\sup_{0\leq\beta\leq\beta_0} Z_{N,\beta} \leq C. \tag{1.4}$$

The proof splits g into a continuous bulk, controlled by Lemma 4.1, and a localized singular part. For the latter, the number of particles in a box makes the effective coupling small enough for Lemma 6.1.

Boxes and the good event. For an integer $J \geq 3$, partition the torus into

$$A_a = J^{-1}(a + [0,1)^{\mathsf{d}}), \qquad a \in (\mathbb{Z}/J\mathbb{Z})^{\mathsf{d}}, \qquad M = J^{\mathsf{d}}. \tag{1.5}$$

Write $L_i = a$ when $X_i \in A_a$, and $n_a = \#\{i : L_i = a\}$. Let $\nu_a = M\mathbf{1}_{A_a}\mu_{\mathrm{unif}}$ be the uniform probability measure in A_a, and let Q_J denote box averaging: $(Q_J u)|_{A_a} = \int u\,\mathrm{d}\nu_a$. These boxes and averages are used in both parts of the proof.

For the singular part, fix $0 < \delta < 1$ and choose the cutoff and the occupancy event together:

$$r = \frac{1}{2J}, \qquad \mathcal{A}_{N,J,\delta} = \{n_a \leq \delta N \text{ for every } a\}, \qquad \mathcal{B}_{N,J,\delta} = \mathcal{A}^{\mathsf{c}}_{N,J,\delta}. \tag{1.6}$$

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The singular kernel constructed below is supported at distance at most r. Nonadjacent boxes are separated by at least $J^{-1} = 2r$, so only the same box or adjacent boxes can interact. Here, adjacency includes shared faces, edges, and corners, with periodic labels: the allowed displacements are $\mathcal{V} = \{-1, 0, 1\}^{\mathsf{d}}$. On $\mathcal{A}_{N,J,\delta}$, one box contains at most δN particles and two boxes at most $2\delta N$; see Figure 1.

The bulk lemma uses the same family of box averages, with a finer grid if needed after r has been fixed. All grids and cutoffs are fixed independently of N.

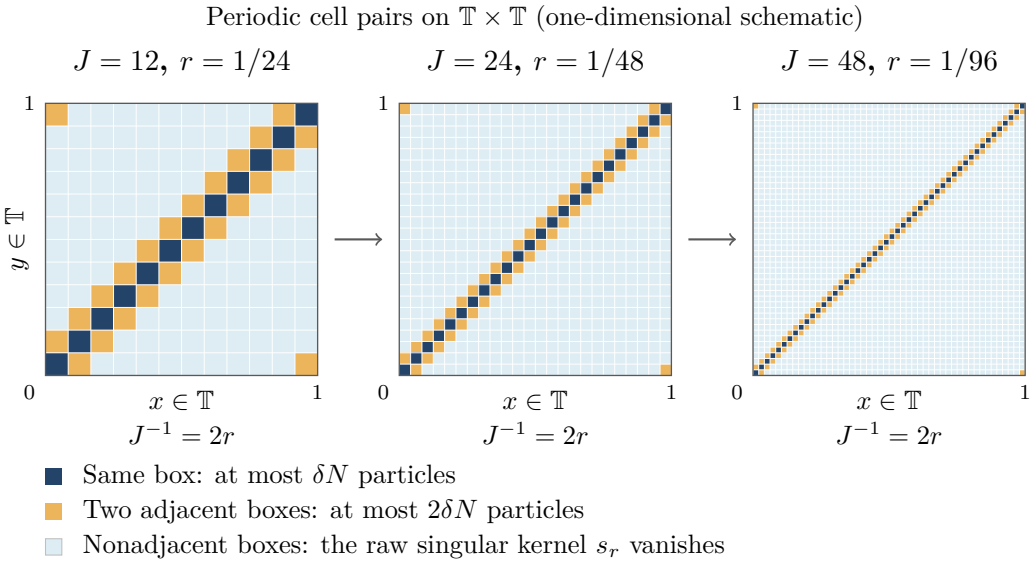


FIGURE 1. The grid is chosen to match the singular cutoff: the cell side is $2r$. In this one-dimensional schematic, $s_r(x - y)$ can be nonzero only in a diagonal square or one of its two neighboring squares, including the periodic corner squares. The colored boxes cover its support; they are not all filled by it. On $\mathcal{A}_{N,J,\delta}$, the one- and two-box counts are bounded by δN and $2\delta N$. Refining the grid and reducing r together narrows the physical band while keeping three possible partners per box (3^{d} in dimension d).

2. Separate the bulk and the diagonal

With $r = (2J)^{-1}$ as above, choose a nonzero, nonnegative, radial $\chi \in C_c^\infty(B_{1/2})$ and set $\phi = \chi * \chi$. Define

$$s_r(x) = \frac{1}{\phi(0)} \sum_{m \in \mathbb{Z}^{\mathsf{d}}} \int_0^r \phi\Big(\frac{x+m}{t}\Big) \frac{\mathrm{d}t}{t}, \qquad c_r = \int s_r \, \mathrm{d}\mu_{\mathrm{unif}}, \qquad \widetilde{s}_r = s_r - c_r, \qquad b_r = \mathsf{g} - \widetilde{s}_r. \tag{2.1}$$

The raw kernel s_r is nonnegative, supported in $\{\mathrm{dist}_{\mathbb{T}^{\mathsf{d}}}(x, 0) \le r\}$, and $c_r = O(r^{\mathsf{d}})$. Since $\widehat{\phi} = |\widehat{\chi}|^2 \ge 0$, radial Fourier inversion gives, for $k \neq 0$,

$$0 \le \widehat{\widetilde{s}}_r(k) = \frac{1}{\phi(0)} \int_0^r t^{\mathsf{d}-1} \widehat{\phi}(tk) \, \mathrm{d}t \le \frac{1}{\phi(0)} \int_0^\infty t^{\mathsf{d}-1} \widehat{\phi}(tk) \, \mathrm{d}t = \widehat{\mathsf{g}}(k). \tag{2.2}$$

Thus both centered parts are positive semidefinite. The Fourier coefficients of b_r, given by the integral from r to infinity, decay rapidly, so b_r is smooth. Its energy is bounded by that of g; hence it inherits (1.3) with the same B.

Group the label vectors L by their occupancies $n = (n_a)$. For a fixed n, retag the particles as $Y_{a,1}, \dots, Y_{a,n_a}$ in box A_a, and let $\mathbb{E}_n$ denote expectation under their independent product law, with each $Y_{a,p}$ distributed according to ν_a. Since the energy is symmetric in the particles,

$$Z_{N,\beta} = \sum_{\substack{n \in \mathbb{N}_0^M \\ \sum_a n_a = N}} \pi_N(n)\, \mathbb{E}_n\big[e^{\frac{\beta}{2}\mathsf{W}_N(\mathsf{g})}\big], \qquad \pi_N(n) = \frac{N!}{M^N \prod_a n_a!}. \tag{2.3}$$

Here, W_N is evaluated on the retagged particles in any order, and the multinomial weights $\pi_N(n)$ sum to one. Split this sum into good vectors, with $\max_a n_a \le \delta N$, and the remaining bad vectors. Fix $0 < \beta_0 < B$ and conjugate $p, q > 1$ with $p\beta_0 < B$. Hölder on the good contribution gives

$$\begin{aligned} Z_{N,\beta} \le \Big(\mathbb{E}\big[e^{\frac{p\beta}{2}\mathsf{W}_N(b_r)}\big]\Big)^{1/p} \Big(\mathbb{E}\big[\mathbf{1}_{\mathcal{A}_{N,J,\delta}} e^{\frac{q\beta}{2}\mathsf{W}_N(\widetilde{s}_r)}\big]\Big)^{1/q} \\ + \mathbb{E}\big[\mathbf{1}_{\mathcal{B}_{N,J,\delta}} e^{\frac{\beta}{2}\mathsf{W}_N(\mathsf{g})}\big], \qquad 0 \le \beta \le \beta_0. \end{aligned} \tag{2.4}$$

The indicator stays on the singular factor, whose coupling $q\beta$ can be large. The next three sections bound the bad event, the bulk, and the singular factor.

3. Discard overcrowded boxes

We first need a rough bound on the full Gibbs weight. For $0 < b < 2\mathsf{d}$, let $A_b = \int e^{b\mathsf{g}/2}\, \mathrm{d}\mu_{\mathrm{unif}} < \infty$. Convexity of the exponential, followed by integration conditional on X_i, gives

$$\begin{aligned} Z_{N,b} &= \mathbb{E}\left[\exp\left\{\frac{1}{N}\sum_{i=1}^{N}\frac{b}{2}\sum_{j\neq i}\mathsf{g}(X_i - X_j)\right\}\right] \\ &\le \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[e^{\frac{b}{2}\sum_{j\neq i}\mathsf{g}(X_i - X_j)}\right] = A_b^{N-1}. \end{aligned} \tag{3.1}$$

This uses only the local logarithmic singularity. Moreover, $A_b \ge 1$ by Jensen, and Hölder gives $Z_{N,\gamma} \le Z_{N,b}^{\gamma/b} \le A_b^{N-1}$ for $0 \le \gamma \le b$.

Each n_a is binomial with parameters N, M^{-1}. For $M^{-1} < \delta$, Chernoff's inequality and a union bound give

$$\begin{aligned} &\mathbb{P}(\mathcal{B}_{N,J,\delta}) \le M e^{-N\mathsf{D}(\delta\|M^{-1})}, \\ &\mathsf{D}(\delta\|M^{-1}) = \delta\log(\delta M) + (1-\delta)\log\frac{1-\delta}{1-M^{-1}} \longrightarrow \infty \quad (J \to \infty). \end{aligned} \tag{3.2}$$

Choose $s > 1$ with $b = s\beta_0 < 2\mathsf{d}$ and $s' = s/(s-1)$. Hölder and (3.1) yield

$$\mathbb{E}\big[\mathbf{1}_{\mathcal{B}_{N,J,\delta}} e^{\frac{\beta}{2}\mathsf{W}_N(\mathsf{g})}\big] \le M^{1/s'} \exp\left\{N\left(\frac{\log A_b}{s} - \frac{\mathsf{D}(\delta\|M^{-1})}{s'}\right)\right\}. \tag{3.3}$$

For fixed $\delta > 0$, choose J large enough to make the exponent negative, then keep this grid fixed as N varies.

4. The continuous bulk

The next lemma records a continuous-kernel entropy–energy partition bound; see Ren and Wang [1, Corollary 4 and Remark 3] for the bounded-interaction torus version. For completeness, Appendix A gives a proof using cell averages, an energy projection, and the estimate of Section 6.

Lemma 4.1 (Continuous kernels controlled by entropy)**.** *Let $K \in C(\mathbb{T}^{\mathsf{d}} \times \mathbb{T}^{\mathsf{d}})$ be real and symmetric, with $\int K(x,y)\, \mathrm{d}\mu_{\mathrm{unif}}(y) = 0$. Suppose its integral operator T_K is nonnegative and, for some $B > 0$,*

$$\mathsf{H}(\rho\mu_{\mathrm{unif}} \mid \mu_{\mathrm{unif}}) \ge \frac{B}{2}\langle \rho - 1, T_K(\rho - 1)\rangle \quad \textit{for every finite-entropy probability density } \rho. \tag{4.1}$$

Then, for every $0 \le \beta_0 < B$,

$$\sup_{N\ge 1}\sup_{0\le\beta\le\beta_0}\mathbb{E}\big[e^{\frac{\beta}{2}\mathsf{W}_N(K)}\big] < \infty. \tag{4.2}$$

For the comparison, set $W_+ = 0$, $W_- = \beta K$, $m_* = \mu_{\mathrm{unif}}$, and $\lambda = 0$ in their entropy hypothesis, with $0 < \varepsilon \le 1/2$ and $(1+\varepsilon)\beta < B$. The diagonal included there contributes at most $\beta\|K\|_\infty/2$ to the exponent.

For $K(x,y) = b_r(x-y)$, the hypotheses follow from (2.2) and log-HLS. In particular,

$$\sup_{N\ge 1}\sup_{0\le\beta\le\beta_0}\mathbb{E}\big[e^{\frac{p\beta}{2}\mathsf{W}_N(b_r)}\big] < \infty \qquad \text{for every fixed } 0 < r < 1/4. \tag{4.3}$$

5. The singular part and the conclusion

The good event makes the singular factor perturbative through the number of particles in each interacting pair of boxes. The moment estimate used below is proved in Lemma 6.1.

Lemma 5.1 (Short-range estimate under a local count bound)**.** *For every* $0 < t_0 < \infty$*, there are* $0 < \delta_0 < 1$ *and an integer* $J_0 \ge 3$ *such that, for each fixed* $0 < \delta \le \delta_0$ *and integer* $J \ge J_0$*, with* $r = (2J)^{-1}$*,*

$$\sup_{N\ge 1}\sup_{0\le t\le t_0}\mathbb{E}\Big[\mathbf{1}_{\mathcal{A}_{N,J,\delta}}e^{\frac{t}{2}\mathsf{W}_N(\widetilde{s}_r)}\Big] < \infty. \tag{5.1}$$

The constant may depend on J, δ, t_0*; these parameters are independent of* N*.*

Proof. Fix $0 \le t \le t_0$. We separate off the box averages, then use (2.3) to estimate the remaining integral for each good occupancy vector.

1. Split the kernel and apply Hölder. Apply the box average Q_J in each variable and write

$$\widetilde{s}_r = a_r + h_r, \qquad h_r = (I - Q_{J,x})(I - Q_{J,y})s_r, \qquad a_r = Q_{J,x}s_r + Q_{J,y}s_r - Q_{J,x}Q_{J,y}s_r - c_r. \tag{5.2}$$

Here, h_r has zero average in each variable within each box. Both a_r and h_r also have zero average in each variable under μ_{unif}. The particle sum splits exactly as

$$\frac{t}{2}\mathsf{W}_N(\widetilde{s}_r) = \frac{t}{N}\sum_{i<j}a_r(X_i,X_j) + \frac{t}{N}\sum_{i<j}h_r(X_i,X_j). \tag{5.3}$$

Hölder with exponents $2, 2$ therefore gives

$$\mathbb{E}\big[\mathbf{1}_{\mathcal{A}_{N,J,\delta}}e^{\frac{t}{2}\mathsf{W}_N(\widetilde{s}_r)}\big] \le \Big(\mathbb{E}\big[e^{t\mathsf{W}_N(a_r)}\big]\Big)^{1/2}\Big(\mathbb{E}\big[\mathbf{1}_{\mathcal{A}_{N,J,\delta}}e^{t\mathsf{W}_N(h_r)}\big]\Big)^{1/2}. \tag{5.4}$$

We bound the first factor by taking r small and the second by taking δ small.

2. The averaging term is perturbative. Every box average of the nonnegative kernel s_r is at most Mc_r. Since $c_r \le Cr^{\mathsf{d}}$ and $Mr^{\mathsf{d}} = 2^{-\mathsf{d}}$, we have

$$\|a_r\|_\infty \le C, \qquad \sup_x\int |a_r(x,y)|\,\mathrm{d}\mu_{\mathrm{unif}}(y) \le 4c_r \le Cr^{\mathsf{d}}. \tag{5.5}$$

The second estimate follows because each averaging term has integral c_r in either variable. These bounds imply $\sup_x\int(e^{|a_r(x,y)|/\kappa} - 1)\,\mathrm{d}\mu_{\mathrm{unif}}(y) \le Cr^{\mathsf{d}}(e^{C/\kappa} - 1)$. Taking $\kappa(r) = C'/\log(1/r)$ with C' large enough makes the right side at most one for small r. Expanding the exponential and using symmetry gives, for every integer $p \ge 1$,

$$\sup_x\int |a_r(x,y)|^p\,\mathrm{d}\mu_{\mathrm{unif}}(y) + \sup_y\int |a_r(x,y)|^p\,\mathrm{d}\mu_{\mathrm{unif}}(x) \le 2\kappa(r)^p p!, \qquad \kappa(r) \longrightarrow 0. \tag{5.6}$$

Apply Lemma 6.1 with $m = N$, $\kappa = \kappa(r)$, and $\theta = 2t$:

$$\mathbb{E}\big[e^{t\mathsf{W}_N(a_r)}\big] = \mathbb{E}\Big[e^{\frac{2t}{N}\sum_{i<j}a_r(X_i,X_j)}\Big] \le \frac{2}{1-(2tC_{\mathrm{u}}\kappa(r))^2} \le \frac{8}{3}, \qquad 0 \le t \le t_0, \tag{5.7}$$

provided $J \ge J_0$, with J_0 chosen so that $2t_0C_{\mathrm{u}}\kappa((2J_0)^{-1}) \le 1/2$.

3. Few particles make the singular term perturbative. Under the box-uniform measures ν_a, there is a constant κ_*, independent of J, such that, for every integer $p \geq 1$,

$$\sup_{y\in A_b}\int_{A_a}|h_r(x,y)|^p\,\mathrm{d}\nu_a(x)+\sup_{x\in A_a}\int_{A_b}|h_r(x,y)|^p\,\mathrm{d}\nu_b(y)\leq 2\kappa_*^p p!,\qquad p\geq 1. \tag{5.8}$$

Indeed, Cauchy–Schwarz gives $0 \leq \phi \leq \phi(0)$ in (2.1), so $s_r(z) \leq (\log(r/\operatorname{dist}_{\mathbb{T}^{\mathsf{d}}}(z,0)))_+$. Thus $M\int s_r^p\,\mathrm{d}\mu_{\mathrm{unif}} \leq M|B_r|p!/\mathsf{d}^p \leq C^p p!$. The averages subtracted to form h_r are bounded by $Mc_r \leq C$, which proves the same estimate for h_r with a larger constant.

The occupancy decomposition gives exactly

$$\mathbb{E}\big[\mathbf{1}_{\mathcal{A}_{N,J,\delta}}e^{t\mathsf{W}_N(h_r)}\big]=\sum_{\substack{n\in\mathbb{N}_0^M,\ \sum_a n_a=N\\ \max_a n_a\leq\delta N}}\pi_N(n)\,\mathbb{E}_n\big[e^{t\mathsf{W}_N(h_r)}\big]. \tag{5.9}$$

Fix one vector in this sum. Its particles form M independent groups, each of size at most δN. Only the same or adjacent boxes interact: s_r vanishes on other box products, and box averaging preserves this property for h_r. Thus each group interacts with at most $K-1$ others, where $K = 3^{\mathsf{d}}$.

Apply the grouped conclusion (6.4) of Lemma 6.1, with $m = N$, $m_* = \max_a n_a$, $\kappa = \kappa_*$, and $u = 2t$ since $\mathsf{W}_N(h_r) = 2N^{-1}\sum_{i<j} h_r(X_i, X_j)$. Choose δ_0 so that $8KC_{\mathrm{u}}\kappa_* t_0\delta_0 \leq 1/2$. Then

$$4KC_{\mathrm{u}}\kappa_*(2t)\frac{m_*}{N}\leq 8KC_{\mathrm{u}}\kappa_* t_0\delta\leq\frac{1}{2},\qquad \mathbb{E}_n\big[e^{t\mathsf{W}_N(h_r)}\big]\leq\left(\frac{8}{3}\right)^M. \tag{5.10}$$

This is uniform over all good occupancy vectors. Their multinomial weights sum to at most one, so (5.9) yields

$$\sup_{N\geq 1}\sup_{0\leq t\leq t_0}\mathbb{E}\big[\mathbf{1}_{\mathcal{A}_{N,J,\delta}}e^{t\mathsf{W}_N(h_r)}\big]\leq\left(\frac{8}{3}\right)^M<\infty. \tag{5.11}$$

Finally, insert this estimate and (5.7) into (5.4). The result is at most $(8/3)^{(M+1)/2}$, uniformly in $N \geq 1$ and $0 \leq t \leq t_0$. Both J_0 and δ_0 are chosen independently of N. □

Conclusion of Theorem 1.1. The case $\beta_0 = 0$ is immediate. Otherwise, choose p, q as in (2.4), and take $\delta \leq \delta_0$ from Lemma 5.1 with $t_0 = q\beta_0$. Next, choose $J \geq J_0$ large enough that $J^{-\mathsf{d}} < \delta$ and (3.3) decays exponentially, and set $r = (2J)^{-1}$. The singular factor is bounded by Lemma 5.1, and the bulk factor by (4.3). Inserting these bounds into (2.4) proves (1.4). □

6. A perturbative estimate for centered arrays

The variables in a local block have different box-uniform laws. The following estimate requires independence and centering, but not identical distributions.

Lemma 6.1 (Canonical-array exponential estimate)**.** *There is a universal $C_{\mathrm{u}} < \infty$ with the following property. Let $m \geq 1$, let $Y_1, \dots, Y_m$ be independent, and let h_{ij}, $i < j$, be real measurable kernels centered in both variables:*

$$\mathbb{E}[h_{ij}(Y_i,y)]=0,\qquad \mathbb{E}[h_{ij}(x,Y_j)]=0\quad \textit{for almost every } y,x. \tag{6.1}$$

Suppose, for some $\kappa > 0$, every integer $p \geq 1$, and every $i < j$,

$$\operatorname*{ess\,sup}_y\mathbb{E}[|h_{ij}(Y_i,y)|^p]+\operatorname*{ess\,sup}_x\mathbb{E}[|h_{ij}(x,Y_j)|^p]\leq 2\kappa^p p!. \tag{6.2}$$

For $S = \sum_{i<j} h_{ij}(Y_i, Y_j)$, every integer $k \geq 1$, and real θ, one has

$$\mathbb{E}[|S|^{2k}]\leq(C_{\mathrm{u}}\kappa m)^{2k}(2k)!,\qquad \mathbb{E}[e^{\theta S/m}]\leq\frac{2}{1-(C_{\mathrm{u}}\kappa|\theta|)^2}\quad \textit{if } C_{\mathrm{u}}\kappa|\theta|<1. \tag{6.3}$$

Zero kernels are allowed. We will use the following grouped form. Partition the m variables into M groups, each containing at most m_* variables. Suppose each group interacts with at most $K-1$

other groups, where $K \geq 1$ is an integer: kernels between all remaining pairs of groups vanish. Then, for $N \geq 1$ and $u \in \mathbb{R}$,

$$\mathbb{E}[e^{uS/N}] \leq \left(\frac{8}{3}\right)^M \quad \text{if} \quad 4KC_{\mathrm{u}}\kappa|u|\frac{m_*}{N} \leq \frac{1}{2}. \tag{6.4}$$

Proof. We give the counting argument. Extend the kernels by $h_{ji}(y,x) = h_{ij}(x,y)$. Independently of the variables Y_i, place each index in I independently with probability $1/2$, and put $D_I = \sum_{i\in I, j\notin I} h_{ij}(Y_i, Y_j)$. Keep the Y_i fixed and average only over the choice of I, denoting this average by $\mathbb{E}_I$. The transpose convention gives

$$\begin{aligned} D_I &= \sum_{i<j} \big(\mathbf{1}_{\{i\in I,\, j\notin I\}} + \mathbf{1}_{\{j\in I,\, i\notin I\}}\big) h_{ij}(Y_i, Y_j), \\ \mathbb{E}_I[D_I] &= \sum_{i<j} \left(\frac{1}{4} + \frac{1}{4}\right) h_{ij}(Y_i, Y_j) = \frac{1}{2} S. \end{aligned} \tag{6.5}$$

Each of the two disjoint ways to separate a pair has probability $(1/2)(1/2) = 1/4$. Thus $S = 2\mathbb{E}_I[D_I]$ for every fixed sample. Jensen's inequality, followed by expectation over the Y_i, now gives

$$\mathbb{E}[|S|^{2k}] \leq 2^{2k}\mathbb{E}_I\mathbb{E}[|D_I|^{2k}]. \tag{6.6}$$

Fix I and expand D_I^{2k}. Each term is indexed by a row word $\mathbf{i} = (i_1, \dots, i_{2k})$ and a column word $\mathbf{j} = (j_1, \dots, j_{2k})$. If any label occurs only once in either word, centering makes the expectation zero. A surviving word therefore uses at most k labels.

For each label $\ell \in \{1, \dots, m\}$, let

$$a_\ell = \#\{v \in \{1, \dots, 2k\} : i_v = \ell\} \tag{6.7}$$

be the multiplicity of ℓ in the row word $\mathbf{i}$, so that $\sum_\ell a_\ell = 2k$. Since every $a_\ell \neq 1$ for a surviving word, the labels with $a_\ell \geq 2$ are the active ones, and there are at most k of them; factors with $a_\ell = 0$ equal 1 in all products and sums over ℓ below. Condition on all column variables. Since every row index lies in I and every column index lies outside I, each factor becomes a function of a single row variable, and the factors sharing the row label ℓ form a group of a_ℓ functions of Y_ℓ. By independence of the row variables, the conditional expectation factors over these groups. Within the group of label ℓ, Hölder with all exponents equal to a_ℓ and the moment hypothesis (6.2) bound the group by $2\kappa^{a_\ell} a_\ell!$, uniformly in the frozen column values. In formulas, write $T = \prod_{v=1}^{2k} h_{i_v j_v}(Y_{i_v}, Y_{j_v})$ and let $\mathcal{C} = \sigma(Y_j : j \notin I)$. The tower property gives $\mathbb{E}[T] = \mathbb{E}\big[\mathbb{E}[T \mid \mathcal{C}]\big]$. The row variables are independent of $\mathcal{C}$ and of one another, so

$$\mathbb{E}[T \mid \mathcal{C}] = \prod_\ell \Phi_\ell(Y_{j_v} : j_v \notin I), \qquad \Phi_\ell(y) = \mathbb{E}\Big[\prod_{v:\, i_v = \ell} h_{\ell j_v}(Y_\ell, y_{j_v})\Big]. \tag{6.8}$$

Hölder with exponents a_ℓ and (6.2) give, for almost every y,

$$|\Phi_\ell(y)| \leq \prod_{v:\, i_v=\ell} \mathbb{E}\big[|h_{\ell j_v}(Y_\ell, y_{j_v})|^{a_\ell}\big]^{1/a_\ell} \leq \big(2\kappa^{a_\ell} a_\ell!\big)^{a_\ell/a_\ell} = 2\kappa^{a_\ell} a_\ell!. \tag{6.9}$$

Since at most k labels survive, $|\mathbb{E}[T]| \leq \mathbb{E}\big[|\mathbb{E}[T \mid \mathcal{C}]|\big] \leq \prod_\ell 2\kappa^{a_\ell} a_\ell!$, that is,

$$\left|\mathbb{E}\left[\prod_{v=1}^{2k} h_{i_v j_v}(Y_{i_v}, Y_{j_v})\right]\right| \leq 2^k \kappa^{2k} \prod_\ell a_\ell!. \tag{6.10}$$

There are $(2k)!/\prod_\ell a_\ell!$ row words with these multiplicities, so their factorials cancel. Write A for the number of surviving multiplicity vectors on at most m labels, and B_k for the number of surviving words. Summing absolute values yields

$$\mathbb{E}[|D_I|^{2k}] \leq 2^k \kappa^{2k} (2k)!\, A B_k. \tag{6.11}$$

It remains to bound AB_k. If $m < 2k$, counting all weak compositions gives $A \le \binom{2k+m-1}{m-1} \le 2^{4k}$, and $B_k \le m^{2k}$. If $m \ge 2k$, choose the $r \le k$ active labels. There are at most 4^k possible positive multiplicity vectors on these labels and at most k^{2k} words. Since $\sum_{r=0}^{k} \binom{m}{r} \le (em/k)^k$, we obtain

$$A \le 4^k (em/k)^k, \qquad B_k \le k^{2k}(em/k)^k, \qquad AB_k \le (4e^2)^k m^{2k}. \tag{6.12}$$

Both cases, together with (6.6)–(6.11), prove the moment bound after enlarging a universal constant.

Finally, $e^x \le 2\cosh x$ and the even-moment bound imply

$$\mathbb{E}[e^{\theta S/m}] \le 2\sum_{k=0}^{\infty} \frac{|\theta|^{2k}\mathbb{E}[|S|^{2k}]}{m^{2k}(2k)!} \le 2\sum_{k=0}^{\infty} (C_{\mathrm{u}}\kappa|\theta|)^{2k}. \tag{6.13}$$

Tonelli justifies the nonnegative series.

For the grouped conclusion, draw an edge between each interacting pair of groups. The graph has degree at most $K-1$. Greedily color its edges with at most $2K-1$ colors so that edges sharing a group have different colors; place all within-group terms in one further class. This writes $S = \sum_{c=1}^{2K} \sum_b S_{c,b}$, allowing empty classes. Within each class, the sums $S_{c,b}$ use disjoint groups and are independent. Each uses at most $2m_*$ variables. Apply the one-block estimate with coefficient $2Ku/N$, then Hölder over the $2K$ classes:

$$\mathbb{E}[e^{uS/N}] \le \prod_{c=1}^{2K} \left(\prod_b \mathbb{E}[e^{2Kuc_{S_{c,b}}/N}] \right)^{1/(2K)} \le \left(\frac{8}{3}\right)^M. \tag{6.14}$$

Indeed, the effective coupling in each factor is at most $4K|u|m_*/N$, and each class has at most M nonempty blocks. This proves (6.4). □

Appendix A. A cell-projection proof of the continuous-kernel bound

We prove Lemma 4.1. The steps are to replace a sample by its histogram and within-cell fluctuation, complete the square in the energy, and estimate the two resulting terms. The histogram gives a finite-dimensional Gaussian integral; continuity makes the remaining centered kernel small.

1. Histogram and energy projection. Use the partitions and box averages of Section 1, choosing the resolution J here for the fixed kernel K. In the application $K = b_r$, the cutoff r stays fixed while this auxiliary grid is refined. With the corresponding labels and counts, set

$$\mu_N = \frac{1}{N}\sum_i \delta_{X_i}, \qquad f_L = \frac{M}{N}\sum_a n_a \mathbf{1}_{A_a}, \qquad \nu_N = \mu_N - f_L\mu_{\mathrm{unif}}. \tag{A.1}$$

Thus $\nu_N(A_a) = 0$. Figure 2 shows this decomposition.

For this proof write $T = T_K$ and $\langle \sigma, T\tau\rangle = \iint K(x,y)\,\mathrm{d}\sigma(x)\,\mathrm{d}\tau(y)$, identifying a density with its measure when needed. Continuity of K makes this form finite on finite signed measures, and positivity extends from densities by approximation. On the mean-zero cell-constant space V_M, define

$$P_M(\nu_N) = \operatorname*{argmin}_{h \in V_M} \langle h - \nu_N, T(h-\nu_N)\rangle. \tag{A.2}$$

If the form has null directions, take the minimizer modulo them. Cauchy–Schwarz for the nonnegative form shows that the mixed functional vanishes on these directions. Projection orthogonality gives

$$\langle h, T(\nu_N - P_M(\nu_N))\rangle = 0 \qquad (h \in V_M). \tag{A.3}$$

Put $z_N = f_L - 1 + P_M(\nu_N)$. Expanding the energy and using (A.3) gives the completed square

$$\begin{aligned} \langle \mu_N - \mu_{\mathrm{unif}}, T(\mu_N - \mu_{\mathrm{unif}})\rangle &= \langle z_N, Tz_N\rangle + \langle \nu_N, T\nu_N\rangle - \langle P_M(\nu_N), TP_M(\nu_N)\rangle \\ &\le \langle z_N, Tz_N\rangle + \langle \nu_N, T\nu_N\rangle. \end{aligned} \tag{A.4}$$

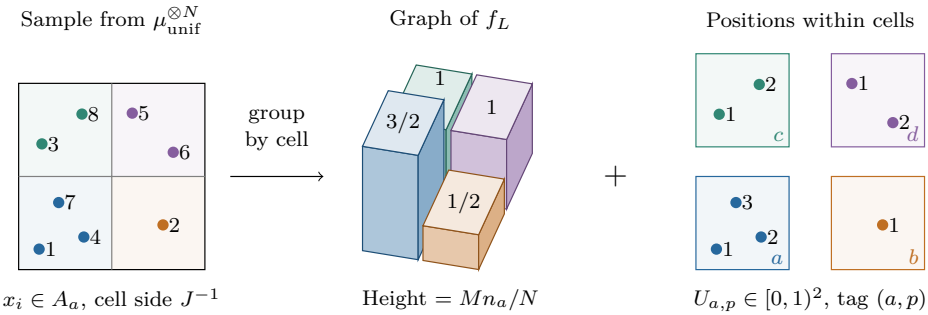


FIGURE 2. A schematic with $M = 4$, $N = 8$. The column heights are the densities Mn_a/N (height 1 is uniform); the offsets retain the positions. Together they recover the sample up to a permutation, via $x_{a,p} = J^{-1}(a + U_{a,p})$. Each box on the right is a copy of the unit square; its points are numbered within that cell. Under the uniform product reference, conditional on the counts these offsets are independent and uniform.

2. Entropy bounds the completed square. For $h \in V_M$, the entropy variational formula and (4.1) imply

$$\begin{aligned} \log \int e^{Th} \, \mathrm{d}\mu_{\text{unif}} &= \sup_{\rho} \{\langle h, T(\rho - 1)\rangle - \mathsf{H}(\rho\mu_{\text{unif}} \mid \mu_{\text{unif}})\} \\ &\le \sup_{\rho} \left\{ \langle h, T(\rho - 1)\rangle - \frac{B}{2}\langle \rho - 1, T(\rho - 1)\rangle \right\} \le \frac{1}{2B}\langle h, Th\rangle. \end{aligned} \tag{A.5}$$

The last step completes the square in $\rho - 1 - h/B$. Also, for every $h \in V_M$,

$$\langle h, Tz_N\rangle = \int Th \, \mathrm{d}\mu_N = \frac{1}{N}\sum_i Th(X_i), \tag{A.6}$$

by the definition of ν_N and projection orthogonality.

Choose coordinates on the nonnull part of V_M, of dimension $q_M \le M - 1$, and let Σ be the matrix of its energy form. For a Gaussian coordinate vector Z with covariance Σ^{-1}, let Z denote the corresponding cell-constant function. The elementary identity $\mathbb{E}_Z[e^{v^\top \Sigma Z}] = e^{v^\top \Sigma v/2}$ and (A.6) give, for $0 \le t < B$,

$$\begin{aligned} \mathbb{E}\big[e^{\frac{tN}{2}\langle z_N, Tz_N\rangle}\big] &= \mathbb{E}_Z\left[\left(\int e^{\sqrt{t/N}\,T\mathsf{Z}} \, \mathrm{d}\mu_{\text{unif}}\right)^N\right] \\ &\le \mathbb{E}_Z\big[e^{\frac{t}{2B}\langle \mathsf{Z}, T\mathsf{Z}\rangle}\big] \\ &= \frac{1}{(2\pi)^{q_M/2}\sqrt{\det(\Sigma^{-1})}} \int_{\mathbb{R}^{q_M}} e^{-\frac{1}{2}(1 - t/B)z^\top \Sigma z} \, \mathrm{d}z \\ &= (1 - t/B)^{-q_M/2}. \end{aligned} \tag{A.7}$$

Here, the inequality is (A.5) applied to $h = \sqrt{t/N}\,\mathsf{Z}$. Tonelli justifies exchanging the Gaussian and particle integrals. The bound is independent of N for the fixed grid.

3. Continuity makes the remainder small. Set

$$r_J = (I - Q_J)_x (I - Q_J)_y K. \tag{A.8}$$

This subtracts the box average in each variable and adds back the joint average. If $\omega_K(\varepsilon)$ is the modulus of continuity when both arguments move by at most ε, then

$$\|r_J\|_\infty \le 4\omega_K(\sqrt{\mathsf{d}}/J) \longrightarrow 0. \tag{A.9}$$

The kernel is centered in each variable under μ_{unif}, and $\langle \nu_N, T\nu_N\rangle = N^{-2}\sum_{i,j} r_J(X_i, X_j)$. Removing equal particle indices in (A.4), we obtain

$$\mathsf{W}_N(K) \le N\langle z_N, Tz_N\rangle + \mathsf{W}_N(r_J) + 5\|K\|_\infty. \tag{A.10}$$

Indeed, the diagonal correction is $N^{-1}\sum_i [r_J(X_i, X_i) - K(X_i, X_i)]$, whose absolute value is at most $5\|K\|_\infty$.

Choose conjugate $p, q > 1$ with $p\beta_0 < B$. By Lemma 6.1, with $\kappa = \|r_J\|_\infty$, a sufficiently fine fixed grid gives $\sup_{N,\, 0\le\beta\le\beta_0} \mathbb{E}[e^{q\beta \mathsf{W}_N(r_J)/2}] \le 8/3$. Hölder and (A.7) now yield

$$\mathbb{E}\big[e^{\frac{\beta}{2}\mathsf{W}_N(K)}\big] \le e^{5\beta_0\|K\|_\infty/2}(1 - p\beta_0/B)^{-q_M/(2p)}(8/3)^{1/q}. \tag{A.11}$$

This proves Lemma 4.1.

For the singular kernel in Section 5, the grid is instead tied to its support by $J^{-1} = 2r$: only adjacent boxes interact, and the particle count makes their effective coupling small.

References

[1] Z. Ren and S. Wang, *Size of chaos for Gibbs measures of mean field interacting diffusions*, Probability Theory and Related Fields (2025). doi:10.1007/s00440-025-01435-z. Accepted version: arXiv:2411.14236v3.